HARAMAYA UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATICS

DEPARTMENT OF INFORMATION SYSTEMS

Course Title	Internet Programming	**Module Code**	InSy-M3092
Module Title	Internet and Web-Technology	**Course Code:**	InSy3091

CP/ECTS: 5 Lecture**: 32** Laboratory**: 48** Tutorial**: 16**

Instructor's Information	Name: Kedir G Email: kedir.genna@haramaya.edu.et
Course Information	Academic Year: **2025/26** Student Batch: **III** Semester: **I**
	Course Schedule: Class Room: **IS-LAB1**
	Mode of Delivery: **Parallel**

Course Description

Overview of internet and the World Wide Web; client-server architecture; web server and security, HTTP protocol; web page design and development; information architecture and visualization; static and dynamic pages. This course deals about client-side scripting languages (JavaScript, VB. Script) and mark up languages (HTML, XML) to design and develop web documents on WWW.Web Development is designed to introduce students to the fundamentals of web site creation concepts. Students will develop, manage, and maintain professional web sites using HTML, XHTML and Cascading Style Sheets. The complete production process including pre-production planning, layout and design considerations (including site maps and navigation), testing, uploading and implementing the site, maintenance and applicable legal and ethical issues are covered.

And also, this course is a continuation of Introduction to Internet programming. It focuses on server-side issues and the construction of medium- to large-scale web-based business-to-consumer (B2C) and business-to-business (B2B) applications using Java Enterprise Technology. In this course, application servers, integration of data from multiple sources, transactions, and delivery of resultant data as XML or WAP to multiple client mechanisms are dealt with. Topics include Java Server Pages (JSP), Servlets, Java Data Base Connectivity (JDBC), Java Naming and Directory Interface (JNDI) and Enterprise Java Beans (EJB). Consideration is also given to dealing with legacy systems. RMI and CORBA are discussed.

Learning Outcomes

- Understand how Internet programming tasks are accomplished.
- Build Internet tools that assist in automating data transfer over the net.
- Demonstrate knowledge of website development concepts and terminology
- Demonstrate knowledge of HTML, XHTML, and CSS.
- Demonstrate a significant ability to plan, design, develop, manage, and maintain HTML-based websites.
- Understand the fundamental aspects of all elements found on web pages.
- Understand and use the principles of user interface design to create user-friendly, easily navigated web pages.
- Design web pages that are accessible to everyone regardless of their physical limitations or geographical location.
- Understand the process of creating content for web pages including text, images, animation, audio, and video elements.
- Create functional JavaScript for use in web pages to add interactive applications to a site.
- Understand and use basic XML for web-based databases and dynamic content.
- Understand and utilize Cascading Style Sheets, and Dynamic HTML in the development of dynamic, interactive websites
- Design, develop, implement and maintain HTML, XHTML, CSS, DHTML, JavaScript-enhanced websites
- Enhance students in understanding of the Java language through their creation of Java Applets for use on web pages.
- Introduce students to programming in the PHP language and reveal its ease of use in the web environment.
- Provide students with an understanding of the Ajax technology for building programs requiring asynchronous web communication.
- Enhance student understanding of the Java language by using Java and the Net beans environment to facilitate web program development.

Course Content

Chapter 1: The Fundamentals

1.1. Internet history, Uses and Services
- 1.1.1. Overview of The World Wide Web (WWW)
- 1.1.2. Client-Server Architectural
- 1.1.3. Uniform Resource Locators (URLs)
- 1.1.4. Domain Name System (DNS)

1.2. TCP/ IP Protocols

1.3. HTTP Request Model

Chapter 2: Introduction to Hyper Text Markup Language (HTML) and XML

2.1 HTML
- 2.1.1. Introduction to Markup Languages
- 2.1.2. Document Object Model (DOM) of HTML
- 2.1.3. Headings, Paragraph and Breaks
- 2.1.4. Formatting Text
- 2.1.5. Working With List
- 2.1.6. Working with Graphics
- 2.1.7. Anchors, URLs and Image Maps
- 2.1.8. Tables
- 2.1.9. HTML Frames
- 2.1.10. HTML Forms
- 2.1.11. ActiveX Objects and Applets
- 2.1.12. Special HTML Elements
- 2.1.13. Page Layout and Design Considerations

2.2 XML
- 2.2.1. Basic XML
- 2.2.2. Document Type Definition
- 2.2.3. XML Schema DOM and Presenting XML
- 2.2.4. XML Parsers and Validation
- 2.2.5. XSL and XSLT Transformation
- 2.2.6. News Feed (RSS and ATOM).

Chapter 3: Cascading Style Sheets (CSS)

- 3.1. CSS Basics
- 3.2. Style Sheet Rules
- 3.3. CSS Selectors
- 3.4. Style Properties
- 3.5. Font and Text properties
- 3.6. Foreground and Background properties
- 3.7. Layout and Positioning Properties
- 3.8. Type of CSS Styles
  - 3.8.1. Inline
  - 3.8.2. Embedded
  - 3.8.3. External
  - 3.8.4. Imported
- 3.9. Style Inheritance
- 3.10. Style Rules Precedence
- 3.11. Style Sheet Layers

Chapter 4: Client-Side Scripting Language

4.1. Introduction

4.2. Client-Side Scripting Using JavaScript
- 4.2.1. Introduction to JavaScript
- 4.2.2. JavaScript Basics
- 4.2.3. Variables, Expression, Control Structure, Array, Functions
- 4.2.4. Handling Events and Exception Handling
- 4.2.5. User Inputs: Form Processing
- 4.2.6. Java script Functions
- 4.2.7. Cookies and Hidden Fields

4.3. Dynamic HTML (DHTML)
- 4.3.1. Introduction
- 4.3.2. Dynamic Styles
- 4.3.3. Dynamic Content
- 4.3.4. Positioning and Animation
- 4.3.5. Filters and Transitions

Chapter 5: Server-Side Scripting Basic

- 5.1. Introduction to server-side scripting
- 5.2. Server-side scripting languages
- 5.3. Use Basic Syntax
- 5.4. Send Data to the Web Browser
- 5.5. Write Comments
- 5.6. Utilize Variables
- 5.7. Manipulate Strings
- 5.8. Manipulate Numbers
- 5.9. Work with constants

Chapter 6: HTML Forms and Server-Side Scripting

- 6.1. Use Conditionals and Operators
- 6.2. Validate Form Data
- 6.3. Send Values to a Script Manually
- 6.4. Work with Forms and arrays of data
- 6.5. Use For and While Loops
- 6.6. Create a Simple Form using PHP
- 6.7. Receive Data from a Form in PHP
- 6.8. Introduction to regular expressions

Chapter 7: File and Database management

7.1. **File**
- 7.1.1. Write to Files
- 7.1.2. Read from Files
- 7.1.3. Create Directories
- 7.1.4. Upload Files
- 7.1.5. Rename and Delete Files and Directories

7.2. **Database management**
- 7.2.1. Connect to an existing Database
- 7.2.2. Send Data to a Database
- 7.2.3. Retrieve Data from a Database
- 7.2.4. Modify Existing Data
- 7.2.5. Remove Existing Data
- 7.2.6. Data base security using server side

Chapter 8: Cookies and Sessions

8.1. Describe the stateless model
8.2. Explain the concepts of maintaining state with sessions
8.3. Create and Read data from sessions
8.4. Putting PHP session IDs in pages
8.5. Create and Read data from Cookies
8.6. Destroy a session
8.7. Maintain session data using Cookies
8.8. Add Parameters to a Cookie
8.9. Delete a Cookie

Chapter 9: Content Management Systems (CMS)

Teaching Strategy

The course will be delivered in the form of lectures, demonstration, student presentations, group discussions, and individual and group project works.

Assessment Criteria

- Participation/Attendances----------**10**
- Tests /Assignments----------**25**
- Project ----------**25**
- Final Examination ----------**40**

Role of Instructor(s)

Delivers lectures, prepares reading assignments and topics for group discussion, prepares projects by discussion with student, gives consultation and advises students on project works and assignments, prepares and evaluates quiz, assignment, midterm and final examination.

Role of Students

Attend lectures, lab session and presentation, work in team on group work, participate in group discussion, discusses with the instructor on topics of interest for project work, delivers and presents project work, attend quiz, midterm and final examination.

Required software and/or hardware

Dreamweaver, PHP and MYSQL, Apache web server

Reference

1. Reading text: - HTML complete ,2nd Ed, Paperback by sybex (pages 1-131, and 481-651)
2. Reading text: - Deitel, Deitel &Nieto, internet & world wide web: How to program, Prentice Hall USA,2002
3. HTML, Black Book, by Holzner
4. JavaScript Bible 6th edition by Danny Goodman and Michael Morrison
5. PHP & MySQL Web development by Luke Welling and Laura Thomson
6. JavaScript for the World Wide Web by Tom Negrino and Dori Smith
7. PHP5 and MySQL Bible by Tim Converse and Joyce Park with Clark Morgan
8. The Internet and Its Protocols: A Comparative Approach (The Morgan Kaufmann Series in Networking) by Adrian Farrel
9. **http://www.w3schools.com/html/default.asp**
10. **http://www.w3schools.com/css/default.asp**
11. Computer Networking with Internet Protocols and Technology by William Stallings
12. Craig D. Knuckles, David S. Yuen, Web Applications: Concepts & Real World Design, John Wiley & Sons, ISBN 0-471-20458-7 (paperback) or 0-471-42929-5.
13. Harvey & Paul, Internet & World Wide Web: How to Program (4th Edition) (How to Program (Deitel)), Deitel & Associates Inc., 2007.
14. Thomas A. Powell, HTML & XHTML: The Complete Reference, McGraw-Hill
15. Neil Gray, Web Server Programming (John Wiely & Sons)
16. Rasmus Lerdorf, Peter MacIntyre, Kevin Tatroe, Programming PHP (2nd Ed.) (O'Reilly)
17. Goodman, Java script bible (3rd Edition), (Gage Publishing)
18. Gundavaram, S. CGI Programming on the World Wide Web, (O'Reilly and Associates Publishing)
19. Horton, William, The Web Page Design Cookbook. (John Wiley & Sons) Ajay Vohra, Deepak Vohra, Pro XML Development with Java Technology (Apress)